An Introduction to Bayesian Computing with INLA

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Agenda of the tutorial

- Introduction
- Latent Gaussian models
- Essentials of the INLA approach
- The `R-INLA` package
- Some examples with the `R-INLA` package

!!! Slides + example html will be available on
`https://sites.google.com/a/r-inla.org/stbook/`
(or write me: `michela.cameletti@unibg.it`)

Introduction:
why do we need INLA?

Bayesian inference

- Very simple idea: $p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$
- Simulation based inference through MCMC starting from the early '90s.
- Popular MCMC software (generic) tools: `WinBUGS`, `JAGS` and `Stan` + specific `R` packages (e.g. `MCMCpack`).
- The issue of Bayesian computing is not completely *solved* even though MCMC is available...

From MCMC to INLA

- MCMC methods are flexible and able to deal with virtually any type of data and model.
- But MCMC methods can be extremely and impractically slow. For this reason the complexity of the model and the database dimension often remain fundamental issues.
- The Integrated Nested Laplace Approximation (INLA) algorithm proposed by [Rue et al., 2009] is a *deterministic* algorithm for Bayesian inference.
- The INLA algorithm is designed for the class of *latent Gaussian models* and compared to MCMC it provides (as) accurate results in a shorter time.
- INLA has become very popular amongst statisticians and applied researchers and in the past few years the number of papers reporting usage and extensions of the INLA method has increased considerably.

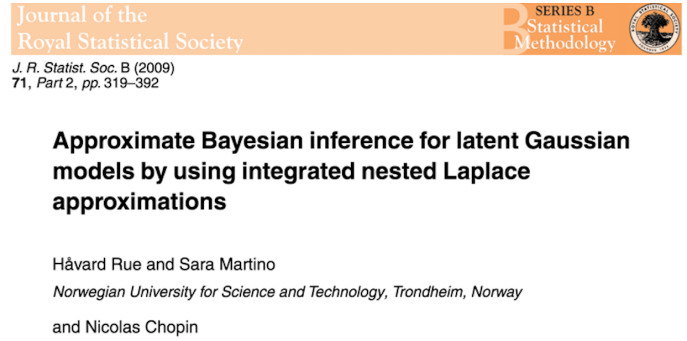
INLA community

http://www.r-inla.org/

- The website contains source code, examples, papers and reports discussing the theory and applications of `INLA`.
- There is also a discussion forum where users can post queries and requests of help.
- Almost each year there is an INLA-related scientific meeting (see `http://www.r-inla.org/scientific-meetings`).

Main references

2009: original paper (745 Citations)


Journal of the Royal Statistical Society SERIES B Statistical Methodology

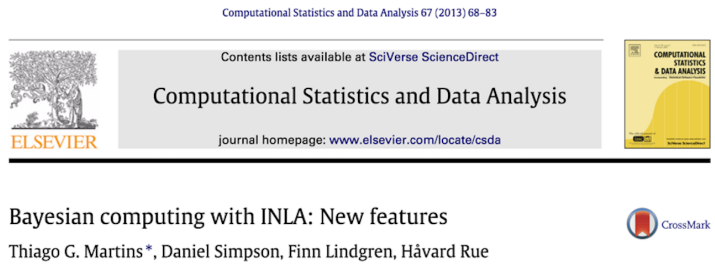
J. R. Statist. Soc. B (2009)
71, Part 2, pp. 319–392

Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations

Håvard Rue and Sara Martino
Norwegian University for Science and Technology, Trondheim, Norway

and Nicolas Chopin

2013: formalisation of new developments in `R-INLA`


Computational Statistics and Data Analysis 67 (2013) 68–83

Contents lists available at SciVerse ScienceDirect

Computational Statistics and Data Analysis

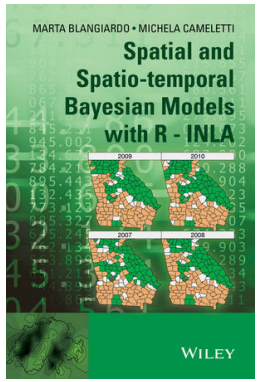
journal homepage: www.elsevier.com/locate/csda

Bayesian computing with INLA: New features

Thiago G. Martins*, Daniel Simpson, Finn Lindgren, Håvard Rue

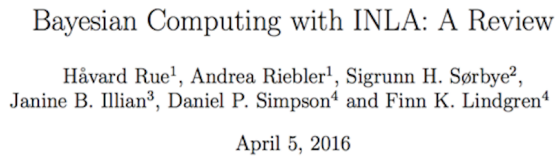
2015: INLA book

`https://sites.google.com/a/r-inla.org/stbook/`



2016: INLA review

`https://arxiv.org/abs/1604.00860`


Bayesian Computing with INLA: A Review

Håvard Rue[1], Andrea Riebler[1], Sigrunn H. Sørbye[2], Janine B. Illian[3], Daniel P. Simpson[4] and Finn K. Lindgren[4]

April 5, 2016

Latent Gaussian models (LGMs)

Latent Gaussian models (LGMs)

- The general problem of (parametric) inference is posited by assuming a probability model for the observed data $\boldsymbol{y} = (y_1, \ldots, y_n)$, as a function of some relevant parameters

$$\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\psi}_1 \sim p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\psi}) = \prod_i p(y_i \mid \theta_i, \boldsymbol{\psi}_1)$$

- Often (in fact for a surprisingly large range of models!), we can assume that the parameters are described by a **Gaussian Markov Random Field** (GMRF)

$$\boldsymbol{\theta} \mid \boldsymbol{\psi}_2 \sim \text{Normal}(\boldsymbol{\mu}(\boldsymbol{\psi}_2), \boldsymbol{Q}^{-1}(\boldsymbol{\psi}_2))$$
$$\theta_i \perp\!\!\!\perp \theta_j \mid \boldsymbol{\theta}_{-i,j} \iff Q_{ij}(\boldsymbol{\psi}_2) = 0$$

  where
  - The precision matrix $\boldsymbol{Q}$ depends on some hyperparameters $\boldsymbol{\psi}_2$.
  - The notation "$-i, j$" indicates all the other elements of the parameters vector, excluding elements i and j.

  !!! The components of $\boldsymbol{\theta}$ are supposed to be *conditionally independent* with the consequence that $\boldsymbol{Q}$ is a sparse precision matrix $\rightsquigarrow$ very efficient numerical algorithms.

LGMs as a general framework

- In general we can re-express a LGM as a three-stage hierarchical model:

$$\begin{aligned} \boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\psi}_1 &\sim \prod_i p(y_i \mid \theta_i, \boldsymbol{\psi}_1) && \text{Data model} \\ \boldsymbol{\theta} \mid \boldsymbol{\psi}_2 &\sim p(\boldsymbol{\theta} \mid \boldsymbol{\psi}_2) = \text{Normal}(\boldsymbol{\mu}(\psi_2), \boldsymbol{Q}^{-1}(\boldsymbol{\psi}_2)) && \text{GMRF prior} \\ \boldsymbol{\psi} = (\boldsymbol{\psi}_1, \boldsymbol{\psi}_2) &\sim p(\boldsymbol{\psi}) && \text{Hyperprior} \end{aligned}$$

The joint posterior is

$$p(\boldsymbol{\theta}, \boldsymbol{\psi} \mid \boldsymbol{y}) \propto p(\boldsymbol{\psi}) p(\boldsymbol{\theta} \mid \boldsymbol{\psi}) \prod_i p(y_i \mid \theta_i, \boldsymbol{\psi})$$

- The dimension of $\boldsymbol{\theta}$ can be very large (eg 10^2-10^5).
- Conversely, the dimension of $\boldsymbol{\psi}$ must be relatively small (typically 2 to 5, but not exceeding 20) to avoid an exponential increase in the computational costs of the model.

Additive models

- A very general way of specifying the problem is specifying a distribution for y_i characterized by a parameter ϕ_i (usually the mean) defined as a function of a structured linear predictor η_i, defined on a suitable scale, such that $g(\phi_i) = \eta_i$ (e.g. logistic for binomial data):

$$\eta_i = \underbrace{\beta_0 + \sum_{m=1}^{M} \beta_m x_{mi}}_{\text{Fixed effects}} + \underbrace{\sum_{l=1}^{L} f_l(z_{li})}_{\text{Random effects}}$$

  where
  - β_0 is the overall intercept;
  - $\boldsymbol{\beta} = \{\beta_1, \ldots, \beta_M\}$ quantify the effect of the fixed covariates $\boldsymbol{x} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_M)$ on the response;
  - $\boldsymbol{f} = \{f_1(\cdot), \ldots, f_L(\cdot)\}$ is a set of functions defined in terms of some covariates/indexes $\boldsymbol{z} = (\boldsymbol{z}_1, \ldots, \boldsymbol{z}_L)$. They represent *specific Gaussian processes* (aka model components). Used to relax linear relationship of covariates, include random effects and temporal and/or spatial dependence.
- The joint distribution

$$\boldsymbol{\theta} = \{\beta_0, \boldsymbol{\beta}, \boldsymbol{f}\} \sim \text{Normal}(\boldsymbol{0}, \boldsymbol{Q}^{-1}(\boldsymbol{\psi})) = \text{GMRF}(\boldsymbol{\psi})$$

LGMs — examples

Upon varying the form of the functions $f_l(\cdot)$, this formulation can accommodate a wide range of models including GLM/GAM/GLMM/GAMM/...:

- Standard regression
  - $f_l(\cdot) = \text{NULL}$
- Smoothing spline
  - $f_l(\cdot) \sim RW$
- Hierarchical models
  - $f_l(\cdot) \sim \text{Normal}(0, \sigma_f^2)$
    $\sigma_f^2 \sim$ some common distribution
- Spatial and spatio-temporal models
  - Areal data: $f_1(\cdot) \sim \text{CAR}$ (Spatially structured effects)
    $f_2(\cdot) \sim \text{Normal}(0, \sigma_{f_2}^2)$ (Unstructured residual)
  - Geostatistical data: $f(\cdot) \sim$ Gaussian field
  - Temporal component: $f(\cdot) \sim \text{AR}(\phi, \sigma_\varepsilon^2)$
- Survival models, logGaussian Cox Processes, etc.

LGMs — examples

The hyperparameters $\boldsymbol{\psi}$ can include:

- likelihood hyperparameters
  - variance of observation noise
  - dispersion parameter in the negative binomial model
  - probability of a zero in zero-inflated models

- parameter/latent field hyperparameters
  - variance of unstructured effects
  - correlation of multivariate effects
  - range and variance of spatial effects
  - autocorrelation parameter

MCMC and LGMs

- (Standard) MCMC methods can perform poorly when applied to (non-trivial) LGMs. This is due to several factors
  - The components of the latent Gaussian field $\boldsymbol{\theta}$ tend to be highly correlated, thus impacting on convergence and autocorrelation
  - Especially when the number of observations is large, $\boldsymbol{\theta}$ and $\boldsymbol{\psi}$ also tend to be highly correlated
  - Time to run can be very long
- Blocking and overparameterisation can **alleviate**, but rarely eliminate the problem

The Integrated Nested Laplace Approximation (INLA) approach

Ingredients

- The first "ingredient" of the INLA approach is the definition of conditional probability, which holds for any pair of variables (x, z) — and, technically, provided $p(z) > 0$

$$p(x \mid z) =: \frac{p(x, z)}{p(z)} \rightarrow p(x, z) = p(x \mid z)p(z)$$

$p(x \mid z)$ can be re-written as

$$p(z) = \frac{p(x, z)}{p(x \mid z)}$$

- In particular, a conditional version can be obtained further considering a third variable w as

$$p(z \mid w) = \frac{p(x, z \mid w)}{p(x \mid z, w)}$$

which is particularly relevant to the Bayesian case.

Ingredients

- The second "ingredient" is **Laplace approximation**.
- Main idea: approximate the integral

$$\int f(x)dx = \int \exp[\log f(x)]dx$$

by means of a Taylor's series expansion around the mode $x^* = \text{argmax}_x \log f(x)$:

$$\int f(x)dx \approx \int \exp\left[\log f(x^*) + \frac{(x-x^*)^2}{2} \left.\frac{\partial^2 \log f(x)}{\partial x^2}\right|_{x=x^*}\right]$$

- Setting $\sigma^{2^*} = -1/\left.\frac{\partial^2 \log f(x)}{\partial x^2}\right|_{x=x^*}$ we can re-write

$$\int f(x)dx \approx f(x^\star) \int \exp\left[-\frac{(x-x^*)^2}{2\sigma^{2^*}}\right] dx$$

- Thus, under LA, $f(x) \approx \text{Normal}(x^*, \sigma^{2^*})$.

Laplace approximation — example

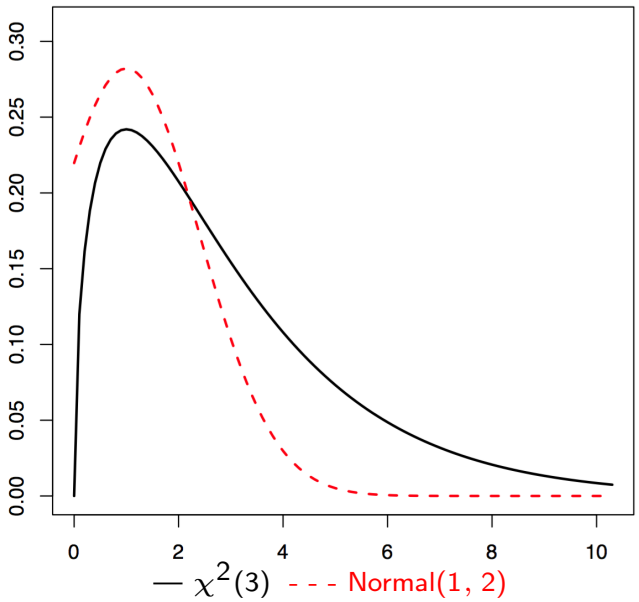
- Consider a χ^2 distribution con k degrees of freedom: $f(x) = \frac{x^{\frac{k}{2}-1} e^{\frac{-x}{2}}}{2^{k/2}\Gamma(k/2)}$
  1. $l(x) = \log f(x) = \left(\frac{k}{2} - 1\right) \log x - \frac{x}{2} + \text{constant}$
  2. $l'(x) = \frac{\partial \log f(x)}{\partial x} = \left(\frac{k}{2} - 1\right) x^{-1} - \frac{1}{2}$
  3. $l''(x) = \frac{\partial^2 \log f(x)}{\partial x^2} = -\left(\frac{k}{2} - 1\right) x^{-2}$

- Then
  - Solving $l'(x) = 0$ we find the mode: $x^* = k - 2$
  - Evaluating $-\frac{1}{l''(x)}$ at the mode gives $\sigma^{2*} = 2(k-2)$

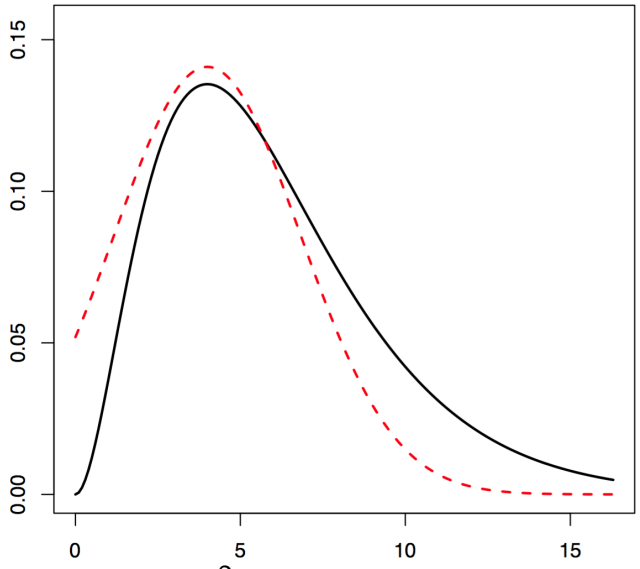
- Consequently, we can approximate $f(x)$ as

$$f(x) \approx \text{Normal}(k-2, 2(k-2))$$

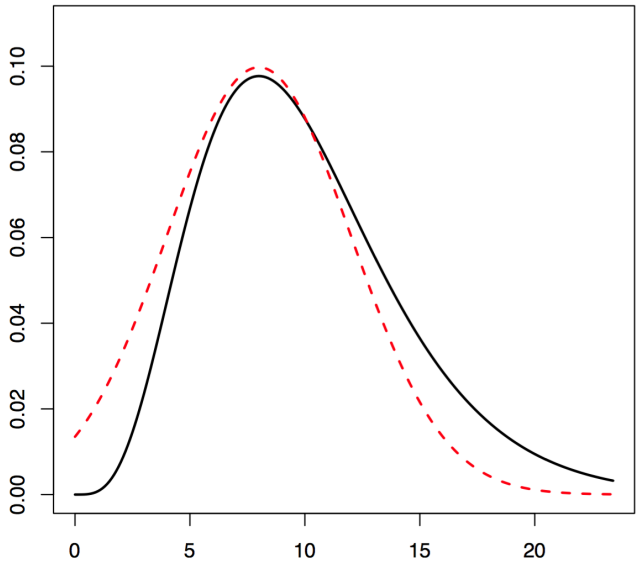
Laplace approximation — example



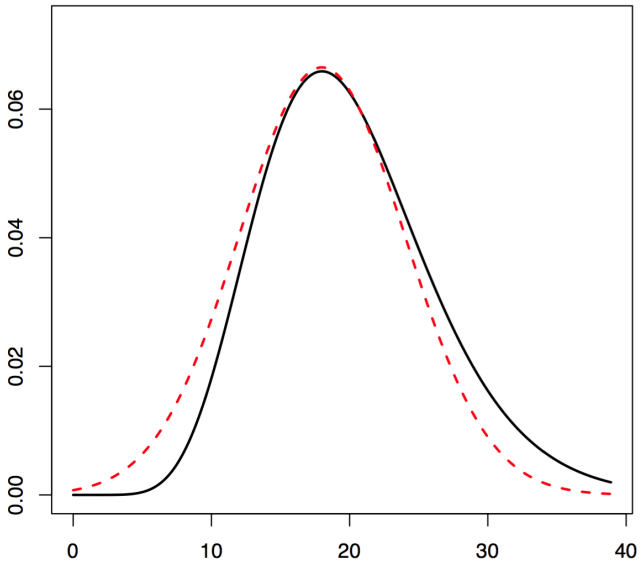
— $\chi^2(3)$ - - - Normal(1, 2)



— $\chi^2(6)$ - - - Normal(4, 8)



— $\chi^2(10)$- - - Normal(8, 16)



— $\chi^2(20)$- - - Normal(18, 36)

Objectives of Bayesian estimation

- The main aim of Bayesian inference for LGMs are the following posterior marginals:

$$
\begin{aligned}
p(\theta_i \mid \boldsymbol{y}) &= \int p(\theta_i, \boldsymbol{\psi} \mid \boldsymbol{y}) d\boldsymbol{\psi} = \int p(\theta_i \mid \boldsymbol{\psi}, \boldsymbol{y}) p(\boldsymbol{\psi} \mid \boldsymbol{y}) d\boldsymbol{\psi} \\
p(\psi_k \mid \boldsymbol{y}) &= \int p(\boldsymbol{\psi} \mid \boldsymbol{y}) d\boldsymbol{\psi}_{-k}
\end{aligned}
$$

- The approximated posterior marginals of interest returned by INLA have the following form:

$$
\begin{aligned}
\tilde{p}(\theta_i \mid \boldsymbol{y}) &= \ldots = \int \tilde{p}(\theta_i \mid \boldsymbol{\psi}, \boldsymbol{y}) \tilde{p}(\boldsymbol{\psi} \mid \boldsymbol{y}) d\boldsymbol{\psi} \\
\tilde{p}(\psi_k \mid \boldsymbol{y}) &= \int \tilde{p}(\boldsymbol{\psi} \mid \boldsymbol{y}) d\boldsymbol{\psi}_{-k}
\end{aligned}
$$

Hyperparameter posterior marginals

$$
\begin{aligned}
p(\boldsymbol{\psi} \mid \boldsymbol{y}) &= \frac{p(\boldsymbol{\theta}, \boldsymbol{\psi} \mid \boldsymbol{y})}{p(\boldsymbol{\theta} \mid \boldsymbol{\psi}, \boldsymbol{y})} \\
&= \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\psi}) p(\boldsymbol{\theta}, \boldsymbol{\psi})}{p(\boldsymbol{y})} \frac{1}{p(\boldsymbol{\theta} \mid \boldsymbol{\psi}, \boldsymbol{y})} \\
&= \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\psi}) p(\boldsymbol{\theta} \mid \boldsymbol{\psi}) p(\boldsymbol{\psi})}{p(\boldsymbol{y})} \frac{1}{p(\boldsymbol{\theta} \mid \boldsymbol{\psi}, \boldsymbol{y})} \\
&\propto \frac{p(\boldsymbol{\psi}) p(\boldsymbol{\theta} \mid \boldsymbol{\psi}) p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\psi})}{p(\boldsymbol{\theta} \mid \boldsymbol{\psi}, \boldsymbol{y})} \\
&\approx \left. \frac{p(\boldsymbol{\psi}) p(\boldsymbol{\theta} \mid \boldsymbol{\psi}) p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\psi})}{\tilde{p}(\boldsymbol{\theta} \mid \boldsymbol{\psi}, \boldsymbol{y})} \right|_{\boldsymbol{\theta} = \boldsymbol{\theta}^*(\boldsymbol{\psi})} =: \tilde{p}(\boldsymbol{\psi} \mid \boldsymbol{y})
\end{aligned}
$$

where

- $\tilde{p}(\boldsymbol{\theta} \mid \boldsymbol{\psi}, \boldsymbol{y})$ is the Laplace approximation of $p(\boldsymbol{\theta} \mid \boldsymbol{\psi}, \boldsymbol{y})$
- $\boldsymbol{\theta} = \boldsymbol{\theta}^*(\boldsymbol{\psi})$ is its mode for a given $\boldsymbol{\psi}$

Since the dimension of $\boldsymbol{\psi}$ is low, we can derive the marginals for $\psi_k \mid \boldsymbol{y}$ directly from the approximation to $\boldsymbol{\psi} \mid \boldsymbol{y}$.

Parameter (or latent field) posterior marginals

The approximation to $p(\theta_i \mid \psi, \boldsymbol{y})$ is slightly more complex, because in general there will be more elements in $\boldsymbol{\theta}$ than there are in ψ and thus this computation is more expensive.

1. **Gaussian Approximation** using $\tilde{p}(\boldsymbol{\theta} \mid \psi, \boldsymbol{y})$.
2. **Full Laplace Approximation**: we can write $\boldsymbol{\theta} = \{\theta_i, \boldsymbol{\theta}_{-i}\}$, use the definition of conditional probability and again Laplace approximation to obtain

$$
\begin{aligned}
p(\theta_i \mid \psi, \boldsymbol{y}) &= \frac{p\left((\theta_i, \boldsymbol{\theta}_{-i}) \mid \psi, \boldsymbol{y}\right)}{p(\boldsymbol{\theta}_{-i} \mid \theta_i, \psi, \boldsymbol{y})} = \frac{p\left((\theta_i, \boldsymbol{\theta}_{-i}), \psi \mid \boldsymbol{y}\right)}{p(\psi \mid \boldsymbol{y})} \frac{1}{p(\boldsymbol{\theta}_{-i} \mid \theta_i, \psi, \boldsymbol{y})} \\
&\propto \frac{p\left(\boldsymbol{\theta}, \psi \mid \boldsymbol{y}\right)}{p(\boldsymbol{\theta}_{-i} \mid \theta_i, \psi, \boldsymbol{y})} \quad \propto \frac{p(\psi)p(\boldsymbol{\theta} \mid \psi)p(\boldsymbol{y} \mid \boldsymbol{\theta}, \psi)}{p(\boldsymbol{\theta}_{-i} \mid \theta_i, \psi, \boldsymbol{y})} \\
&\approx \left. \frac{p(\psi)p(\boldsymbol{\theta} \mid \psi)p(\boldsymbol{y} \mid \boldsymbol{\theta}, \psi)}{\tilde{p}(\boldsymbol{\theta}_{-i} \mid \theta_i, \psi, \boldsymbol{y})} \right|_{\boldsymbol{\theta}_{-i} = \boldsymbol{\theta}^*_{-i}(\theta_i, \psi)} =: \tilde{p}(\theta_i \mid \psi, \boldsymbol{y})
\end{aligned}
$$

This approximation works generally well but is computationally expensive.

Parameter (or latent field) posterior marginals

3. The most efficient algorithm is the **Simplified Laplace Approximation**:

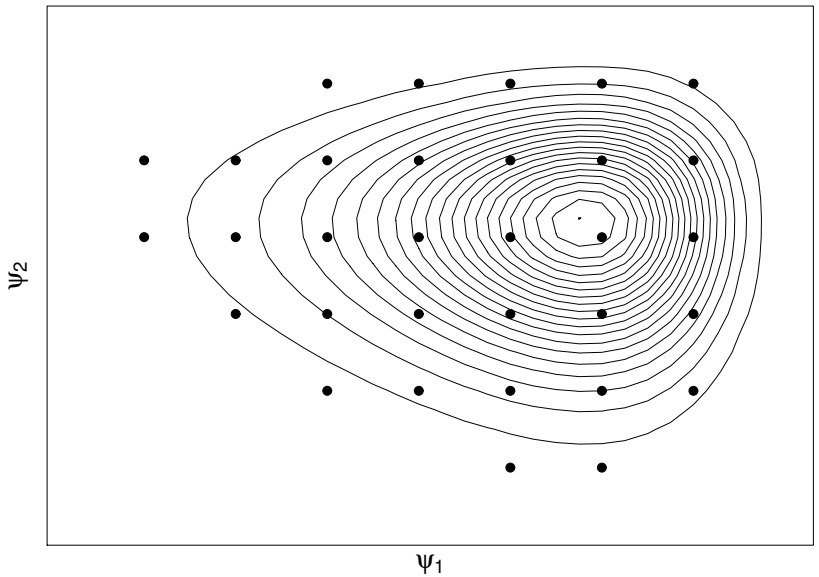
   - based on a Taylor series expansion up to third order of the numerator and denominator of $\tilde{p}(\theta_i \mid \boldsymbol{\psi}, \boldsymbol{y})$.

   - This corrects the Gaussian approximation for error in location and lack of skewness.

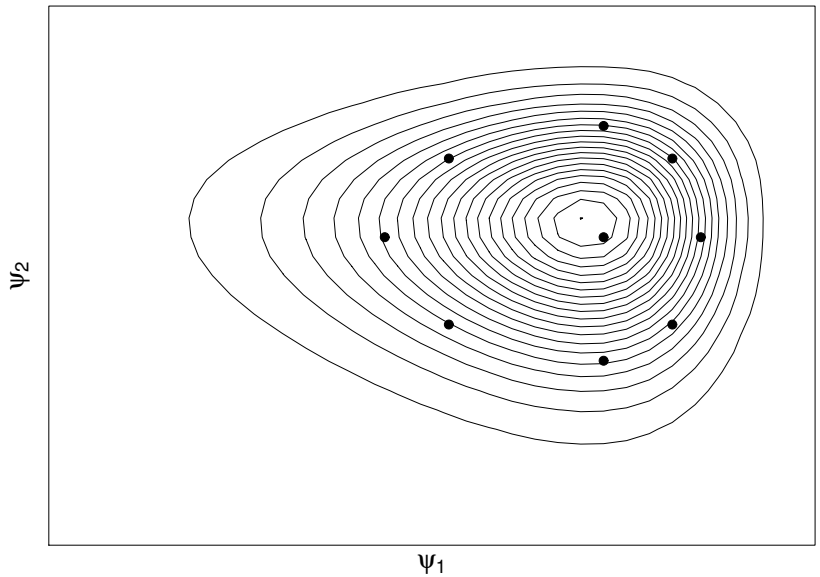
   - This is the algorithm implemented by default by `R-INLA`, but this choice can be modified.
     - If extra precision is required, it is possible to run the full Laplace approximation — of course at the expense of running time!

The INLA steps

Step 1. Explore the joint posterior for the hyperparameters $\tilde{p}(\boldsymbol{\psi} \mid \boldsymbol{y})$ and produce a grid of "good" integration points $\{\boldsymbol{\psi}^\star\}$ associated with the bulk of the mass, together with a corresponding set of area weights $\{\Delta^\star\}$:



Grid strategy



Central Composite Design strategy (CCD)

The CCD strategy is the default one in R-INLA: it produces a lower number of points which are however enough to capture the variability of the joint distribution (see [Martins et al., 2013]).

The INLA steps

Step 2. After the grid exploration, obtain the marginal posterior $\tilde{p}(\psi_k \mid \boldsymbol{y})$ using an interpolation algorithm based on the values of the density $\tilde{p}(\boldsymbol{\psi} \mid \boldsymbol{y})$ evaluated in the integration points $\{\boldsymbol{\psi}^\star\}$ (see Martins et al., 2013).

Step 3. For each integration point in $\boldsymbol{\psi}^\star$ and parameter θ_i, evaluate the approximated marginal $\tilde{p}(\theta_i \mid \boldsymbol{\psi}^\star, \boldsymbol{y})$ for some selected values of θ_i.

Step 4. For each i obtain the marginal posteriors $\tilde{p}(\theta_i \mid \boldsymbol{y})$ using **numerical integration**[1]

$$\tilde{p}(\theta_i \mid \boldsymbol{y}) \approx \sum_{\boldsymbol{\psi}^\star} \tilde{p}(\theta_i \mid \boldsymbol{\psi}^\star, \boldsymbol{y}) \tilde{p}(\boldsymbol{\psi}^\star \mid \boldsymbol{y}) \, \Delta^\star$$

[1] Recall that $p(\theta_i \mid \boldsymbol{y}) = \int p(\theta_i, \boldsymbol{\psi} \mid \boldsymbol{y}) d\boldsymbol{\psi} = \int p(\theta_i \mid \boldsymbol{\psi}, \boldsymbol{y}) p(\boldsymbol{\psi} \mid \boldsymbol{y}) d\boldsymbol{\psi}$

Integrated Nested Laplace Approximation (INLA)

So, it's all in the name...

Integrated Nested Laplace Approximation.

- Because Laplace approximation is the basis to estimate the unknown distributions
- Because the Laplace approximations are nested within one another (since $\tilde{p}(\psi \mid \boldsymbol{y})$ is needed to estimate $\tilde{p}(\theta_i \mid \boldsymbol{y})$).
- Because the required marginal posterior distributions are obtained by (numerical) integration.

The R-INLA package

The INLA package for R

Good news is that all the procedures needed to perform INLA are implemented in a R package. This is effectively made by two components

1. The GMRFLib library
   - A C library for fast and exact simulation of GMRFs
2. The inla program
   - A standalone C program build upon the GMRFLib library (it performs the relevant computation and returns the results in a standardised way)

NB: Because the package R-INLA relies on a standalone C program (and other reasons...), it is not available directly from CRAN.

R-INLA runs natively under Linux, Windows and Mac and it is possible to do multi-threading using OpenMP.

The INLA package for R — Installation

- From R, installation of the stable version is performed typing

```
install.packages("INLA",
repos="http://www.math.ntnu.no/inla/R/stable")
```

- Later, you can upgrade the package by typing

```
library(INLA)
inla.upgrade()
```

- A test-version (which may contain unstable updates/new functions) can be obtained by typing

```
inla.upgrade(testing=TRUE)
```

- Type `inla.version()` to find out the installed version

From the linear predictor to the `formula`

- Recall that the core element of each LGM is the linear predictor

$$\eta_i = \underbrace{\beta_0 + \sum_{m=1}^{M} \beta_m x_{mi}}_{\text{Fixed effects}} + \underbrace{\sum_{l=1}^{L} f_l(z_{li})}_{\text{Random effects}}$$

  related to the observation through the link function: $g(\phi_i) = \eta_i$.

In `R-INLA`:

- The likelihood and the link function $g(\cdot)$ are specified through the `inla(...)` function.
- The linear predictor η_i is specified through a `formula` interface (similarly to the calls to functions such as `lm` and `glm`).

From the linear predictor to the `formula`

For example the following simple linear predictor

$$\eta_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + f(z_i)$$

where

- $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2) \sim \text{Normal}(0, \text{high variance})$ are fixed effects
- $f \sim \text{Normal}(\mathbf{0}, \boldsymbol{Q}_f^{-1}(\tau))$ is used to include a random effect (which depends on the index $\boldsymbol{z}$)

This model is translated in a INLA formula as follows:

```
formula = y ~ 1+ x1 + x2 + f(name=z, model="...")
```

where `y,x1,x2,z` are the variable names in the dataset.

It is possible to include in the `formula` several `f(...)` terms specifying them separately, e.g.

```
formula = y ~ x1 + x2 + f(z1,...)  + f(z2,...)  + f(z3,...)
```

More about the `formula`

- The `f(...)` function can account for several random effects. We have for example:
  - `model="iid"`: independent/exchangeable random effects
  - `model="rw1"` (or `"rw2"`, `"ar1"`): smooth effect of covariates or time effects
  - `model="besag"`: spatially structured effects (CAR)
  - `model="generic"`: user-defined precision matrix
- Type `names(inla.models()$latent)` for the complete list and find descriptions at `http://www.r-inla.org/models/latent-models`
- Several options are available for `f(...)` function (see `?f`):

```
f(name=..., model="...", hyper=...,constr=FALSE,...)
```

  - `hyper`: to specify the prior on the hyperparameters
  - `constr`: sum-to-zero constraint?

Run INLA!

- Call the function `inla(...)` to fit the model, specifying the data, the formula and the likelihood:

```
output=inla(formula, family="...", data=data.frame(y,x1,x2,z))
```

- The data need to be included in a suitable `data.frame` (or `list`).
- The distribution of the data (i.e. the likelihood) is specified with the `family` option.
  - Type `names(inla.models()$likelihood)` for the complete list of likelihood function (we have `poisson`, `binomial`, `gamma`, `beta`, `gaussian` and many others).
  - Visit `http://www.r-inla.org/models/likelihoods` for the complete description.

Options of the `inla(...)` function

The `control.xxx=list(...)` statements in the `inla(...)` function control various part of the INLA program:

- `control.compute`: for computing measures of fit (eg DIC)
- `control.predictor`: for obtaining marginals for the linear predictor / specifying the link function
- `control.family`: for changing the prior distribution of the likelihood hyperparameters
- `control.fixed`: for changing the prior distribution of the fixed effects
- `control.inla`: for changing the strategy to use for the approximations ('gaussian', 'simplified.laplace' (default) or 'laplace') or the grid exploration strategy ('ccd', 'grid')
- ...

Hierarchical models

What are hierarchical models?

'Hierarchical model' is a very broad term that refers to wide range of model set-ups: multilevel models, random effects models, random coefficient models, mixed effect models, ...

- Modelling data with a complex structure
  - large range of structures that can be handled routinely using hierarchical models, e.g. patients nested in hospitals, towns nested in regions, etc.
- Modelling heterogeneity
  - standard regression averages, while hierarchical models additionally model variances
- Modelling dependent data
  - potentially complex dependencies in the outcome over time/space
- Modelling contextuality: micro and macro relations
  - relationship between outcome and confounders depends on individual as well as area level characteristics

A motivating example: Disease mapping

- Structure: Counts of disease cases in a set of small areas
- Question: Which areas have particularly high or low disease rates?
- Question: Can we explain some of the variation in disease rates by area-level covariates?

Example: observed cases of lip cancer y_i diagnosed in Scotland in 1975-1980 at county level $i = 1, \ldots, 56$ areas: expected number of cases E_i are also available using age/sex standardised reference rates and population counts.

- Assume a Poisson likelihood for the disease counts in each area:

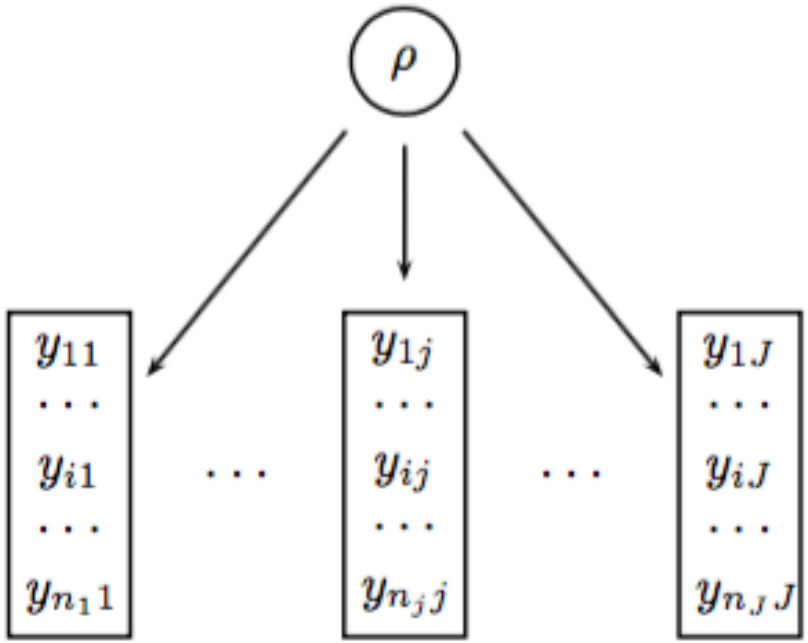
$$y_i \sim \text{Poisson}(\lambda_i) \qquad \lambda_i = \rho_i E_i \qquad i = 1, \ldots, 56$$

- We have 56 relative risk parameters ρ_i (one for each area)
- What prior on ρ_i?

Model assumption 1: pooling

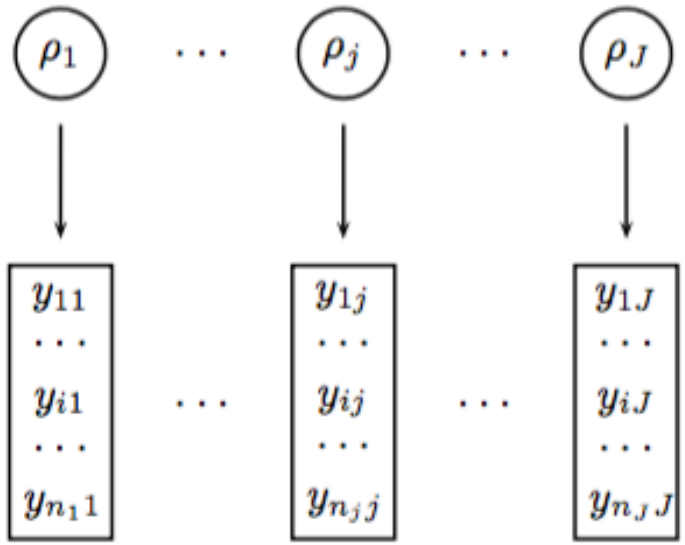
- Assume $\rho_i = \rho \rightsquigarrow$ all the data can be pooled and the individual areas ignored.
- Assume a prior $\rho \sim \text{Gamma}(1, 1)$ $\rightsquigarrow$ conjugate prior.

- One parameter generates all the observations.
- Very easy to implement as it is conjugate (no need for INLA) and all the data are pooled to produce one estimate of the parameter of interest.
- Can be unrealistic (it does not take into account differences in the areas).



Model assumption 2: independent parameters

- All the ρ_i are unrelated, meaning that the areas are analysed independently
- Assume a prior
  $\rho_i \sim \text{Gamma}(1,1); \qquad i = 1, \ldots, 56$
  $\rightsquigarrow$ individual estimates of ρ_i are likely to be highly variable (unless very large sample sizes)



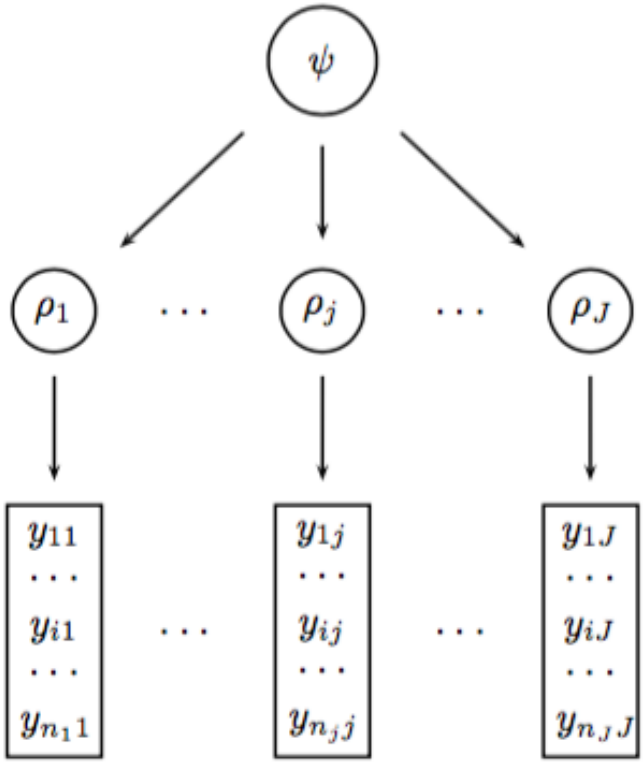
- Every area is treated separately (no exchange of information between these). Estimates close to SMR ($\rho_i \approx y_i/E_i$).
- Again no need for INLA, conjugacy can be exploited.

Model assumption 3: similar (exchangeable) parameters

- All the ρ_i are assumed to be *similar* $\rightsquigarrow$ they come from the same distribution (are generated by the same parameters)
- Assume a hierarchical prior

$$\rho_i \sim \text{Gamma}(a, b)$$

where a and b are unknown parameters and need to be estimated.



- Allow the exchange of information between different levels as they are all connected to each other.
- Assign hyperprior distribution to a and b. How to specify a suitable hyperprior?

A more flexible hierarchical prior for the relative risks

- A gamma random effect prior for the ρ_i is mathematically convenient, but might be restrictive:
  - Covariate adjustment is difficult
  - Not possible to allow for spatial correlation between risks in nearby areas
- A Normal random effect prior on the $\log \rho_i$ is more flexible:

$$\begin{aligned} y_i &\sim \text{Poisson}(\lambda_i = \rho_i E_i) \\ \eta_i = \log(\rho_i) &= b_0 + v_{0i} + b_1 x_i \\ v_{0i} &\sim \text{Normal}(0, \sigma^2_{v0}) \end{aligned}$$

- Need to specify prior distributions for:
  - σ^2_{v0} (between-area variance), e.g. $1/\sigma^2_{v0} \sim \text{Gamma}(0.01, 0.01)$
  - b_0 (mean log relative risk), e.g. $b_0 \sim \text{Normal}(0, 10^6)$
  - b_1 (covariate effect), e.g. $b_1 \sim \text{Normal}(0, 10^6)$
- Advantages:
  - *'borrows strength'* from the likelihood contributions of *all* the areas, via their joint influence on the estimate of the unknown population parameter σ^2_{v0}
  - leads to *global smoothing* of the area RR towards a global mean

Relevant outputs for hierarchical model

- $\exp(b_1)$: increase/decrease in the risk of lip cancer when the covariate x increases of 1 unit
- $\exp(v_{0i})$: (residual) relative risk in area i interpreted with respect to the average relative risk of the whole study region given by $\exp(b_0)$.
- $p(\exp(v_{0i}) > 1 \mid \boldsymbol{y})$ to detect excess risk
- The random effect quantile ratios to quantify their variability. We consider the 5th and 95th quantiles of the area log relative risk distribution:
  - let $q_{5\%}$: 5th quantile of $\boldsymbol{v}_0$ (across areas)
  - let $q_{95\%}$: 95th quantile of $\boldsymbol{v}_0$ (across areas)
  - then $\text{QR}_{90} = \exp(q_{95\%} - q_{5\%})$ = relative risk between the top and bottom 5% of areas
  - We will compute QR_{90} through simulations
- The (between-area) variance of the random effects σ^2_{v0} reflects the amount of extra-Poisson variation in the data.

More and more with INLA (not today...)

Advanced use of INLA

- Multiple likelihood models: rewrite the response variable as a matrix (or list) where the number of columns (or elements in the list) is equal to the number of different likelihood functions. Then in the `inla(...)` call you will have something simular to `family = c("binomial", "Poisson","...")`.
- The `replicate` option can be used to define models where a model component `f(...)` can be replicated giving rise to iid replications with shared hyperparameters:
  `f(time,model="ar1",replicate=person)`
- The `group` option can be used to create dependent replications with a separable correlation structure:
  `f(space,model=spde,group=time,control.group=list(model="ar1"))`
- The `copy` feature is used when a parameter is needed more than once in the model formulation (each linear prediction η_i is connected to more than one element of a model component `f(...)`). For example: $\eta_i = a_i + b_i z_i$ and $(a_i, b_i) \overset{\text{iid}}{\sim} \mathrm{N}_2(\mathbf{0}, \boldsymbol{Q}^{-1})$

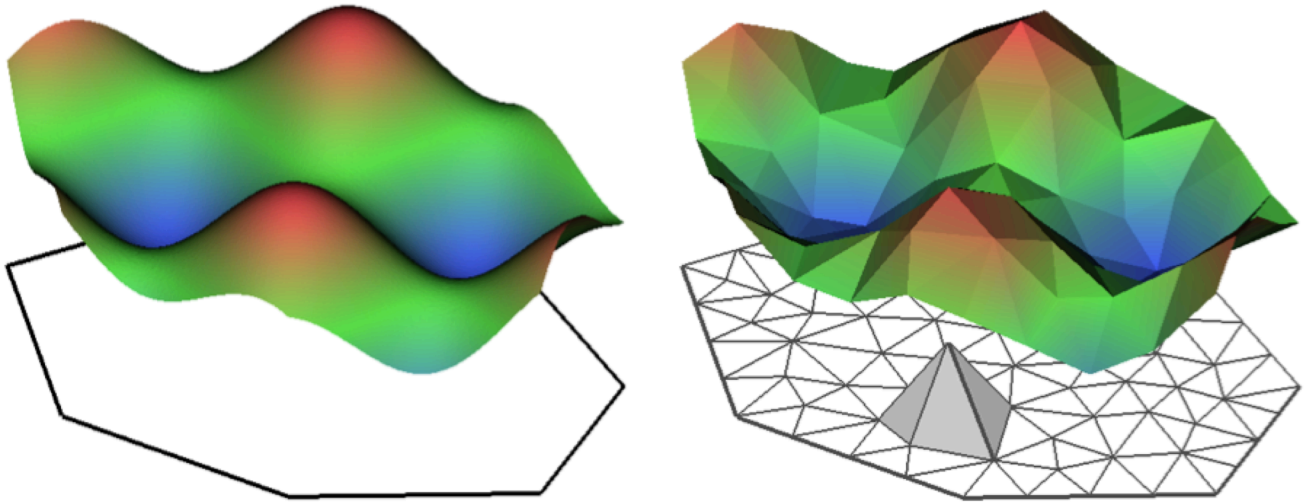
```
formula = y ~ f(i, model = "iid2d", n = 2*n) +
                f(j, z, copy = "i")
```

Spatial (and spatio-temporal) models for areal data

- In the context of hierarchical model it is possible to take into account the spatial dependence amongst the areas (disease mapping), i.e. local smoothing.
- Through conditional autoregressive distributions (CAR models) we specify the distribution of each random effect as if we knew the values of the spatial random effects in neighbouring areas $\rightsquigarrow$ borrowing information from neighbouring areas
- BYM is the most used method for disease mapping
  `formula = ... + f(areaID, model="bym", graph=...)`
- It is also possible to include temporal dependence + space-time model component (through the `group` option).

Spatial (and spatio-temporal) models for geostatistical data

- The common framework to geostatistical models is that of Gaussian fields (or processes) which are based on the Multivariate Normal distribution.
- Main aim: To reconstruct a latent spatial field from a finite set of noisy observations taken at a finite number of spatial locations
- The stochastic partial differential equation (SPDE) approach proposed by [Lindgren et al., 2011] represents a continuous spatial process with Matèrn covariance function using a discretely indexed spatial random process (ie a GMRF), which is characterized by a sparse precision matrix and enjoys computational benefits in terms of fast inference.



Summary

- INLA is a fast method to do Bayesian inference with latent Gaussian models and INLA is an R-package that implements this method with a flexible and simple interface
- The INLA approach is not a rival/competitor/replacement to/of MCMC, just a better option for the class of LGMs.
- The basic idea behind the INLA procedure is simple and it is very easy to use for simple problems, but for more complex problems it may happen that you have to struggle a bit.
- Ask for help to the INLA team through the `r-inla.org` discussion forum or through e-mail (see `inla.version()`) for a list of e-mails.

INLA is still an active project

- A big push into including more general likelihoods and larger problems
- Possibility to pass a user-define model component `f(...)` through the `inla.rgeneric.define(...)` function
- New features for detecting prior sensitivity and specifying priors for parameters (PC priors, see `http://www.r-inla.org/examples/case-studies/pc-priors-martins-et-al-2014`)
- Extensions beyond the basic LGM framework

Acknoledgments

- Marta Blangiardo, Imperial college London





- All the INLA group (Finn & Haavard, especially)

... YOU!

For your attention today!